

tiles can sense, react and acclimatise their condition(s) to given external weather circumstances. These follow the concept of artificial intelligence (AI)—these operate like a brain, and have the abilities of cognition, reasoning and activating. Though embedded with advanced processing technology, these are worn comfortably like normal daily clothing.

Smart fabrics were initially developed for performance-enhancing purpose for professional athletes, but now their applications are expanding across various industries.

**Sports and fitness.** New technology advances in sportswear make it possible for players to monitor their activities and better their performances. Smart garments capture their body temperature, breathing patterns and other physiological parameters. These can also track the number of steps and distances covered, and check performance records.

Besides, smart fabrics can actively stimulate muscles by using electrical muscle stimulation. These can also protect from many possible injuries and health hazards.

**Health.** In the healthcare domain, devices are being embedded into blan-

kets, shirts and other essential tools. Data from the devices allows tracking of physiological status of patients efficiently. The biggest advantage of wireless, sensors-embedded garments is that these can remotely monitor the health of terminally-ill patients.

Also, these reduce the number of visits to healthcare centres as the devices themselves provide updates on the health status of patients.

**Military and defence.** Smart clothing helps increase the safety and effectiveness of military forces. In emergency cases, smart clothing can be extremely useful and quick to send out alerts.

For the safety and protection of military personnel on duty in harsh weather conditions, real-time information is required, and smart clothing equipped with communication and weather-resistant features have proven to be effective.

In addition, these textiles have colour-changing properties to match the environment. These can send vital signs in case of any mishaps. In response, the concerned unit can conduct a remote monitoring of the situation and fix the problems.

At present, uptake of smart cloth-

ing is still witnessing a sluggish pace in comparison with other wearable products. High price is one of the restraining factors in the successful commercialisation of smart fabrics.

Consequently, their popularity among regular consumers is yet to be achieved. And most smart clothing products do not look very attractive, which results in their low consumption. Awareness about the USP of smart garments is still weak to strike a positive impact on consumers.

Another aspect crippling the growth of the e-textile domain is lack of development in terms of universal standards and IP protection for smart textiles. Therefore stakeholders are hesitant to scale up their business in the absence of standardisation.

There are various other technical challenges. Integration of numerous sensors gives out hordes of information, which makes it difficult to classify.

The fluctuating nature of data questions the accuracy and reliability of output information. Therefore more focus on research in addition to involvement from manufacturers and stakeholders can address the challenges plaguing the domain.

—Potshangbam July

## 3 SMART INHALERS ENABLING BETTER RESPIRATORY CARE

**Smart inhalers can be used to administer the correct dose of medicine and track patient health when connected to an app that can gather data**

**R**espiratory diseases like asthma and chronic obstructive pulmonary disease (COPD) are on the rise due to such factors as air pollution, active or passive smoking, exposure to certain chemicals or fumes, etc. According to World Health Organization, COPD will become the third leading cause of death worldwide by 2030. This drives the market for digital respiratory devices, which include both therapeutic (inhalers, nebulisers) and diagnostic devices.

When it comes to traditional inhalers, there is no way of determining if they are being properly used and which

prescription is suitable for a particular patient. The type of standard inhaler used varies according to the intensity of the respiratory disorder and the need for its diagnosis. Smart inhalers, on the other hand, can be used to administer the correct dose of medicine and track patient health when connected to an app that can gather data.

According to a report by The Insight Partners, the global smart inhalers market is predicted to grow at a CAGR of 25.5 per cent during the forecast period of 2019 to 2027, to reach US\$ 8033.21 million by 2027. Well-known companies like Teva Pharmaceutical Industries, Glenmark, 3M, Pneuma

Respiratory, AstraZeneca, Propeller Health, Adherium and Amiko Digital Health are among the leading organisations in this field. Glenmark, headquartered in Mumbai, launched the first digital inhaler in 2016.

UK-based Amiko, founded by Alessandro Fato, Alexander Marinov, Duilio Macchi and Martijn Grinovero in 2015, is a developer of advanced medication sensor technologies and artificial intelligence (AI)-powered digital health solutions. Their product Respiro is an end-to-end digital medicine solution that uses advanced smart inhalers and software to measure and improve outcomes for respiratory diseases.